

Network Layer: Logical Addressing

IPv4 ADDRESSES

An **IPv4 address** is a **32-bit** address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.

The identifier used in the network layer to identify the connection of each device to the internet is called Internet address or IP address.



Note

An IPv4 address is 32 bits long.

**The IPv4 addresses are unique
and universal.**



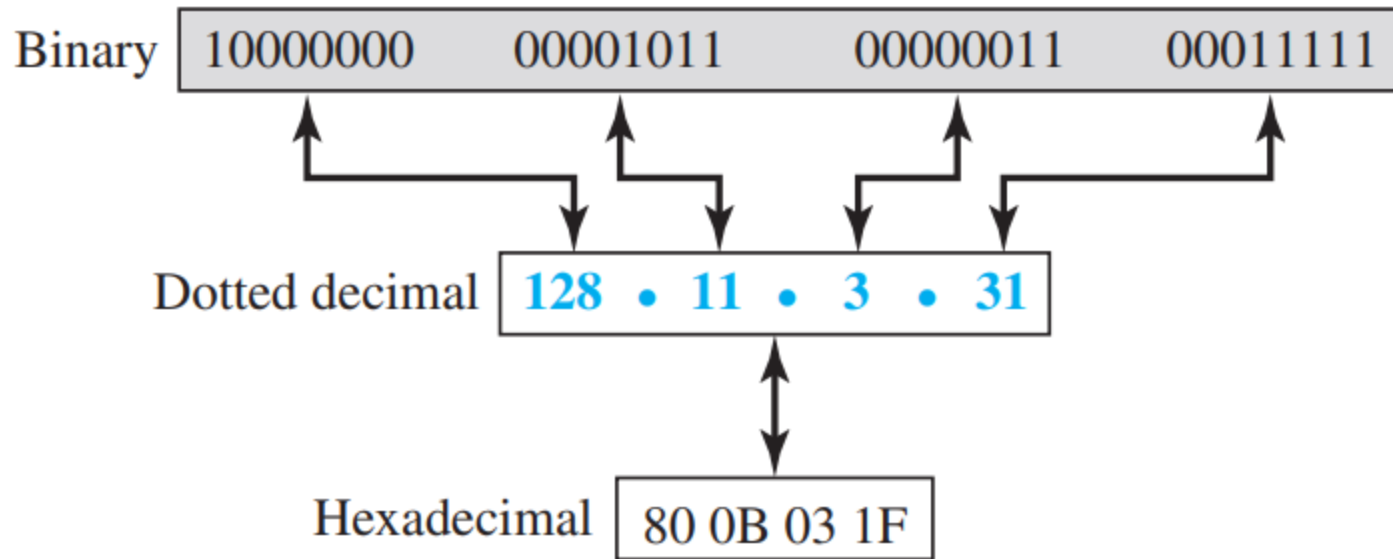
Note

**The address space of IPv4 is
 2^{32} or 4,294,967,296.**

Address Space

- A Protocol like IPv4 that defines addresses has an address space.
- Total number of addresses used by the protocol
- If a protocol uses b bits to define an address, the address space is 2^b because each bit has two different values (0 or 1)
- IPv4 uses 32 bit addresses, which means the address space is 2^{32} or 4,294,967,296 (more than four billion)

Dotted-decimal notation, binary notation and Hexadecimal notation

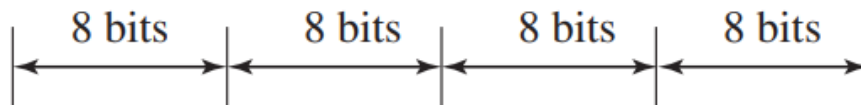
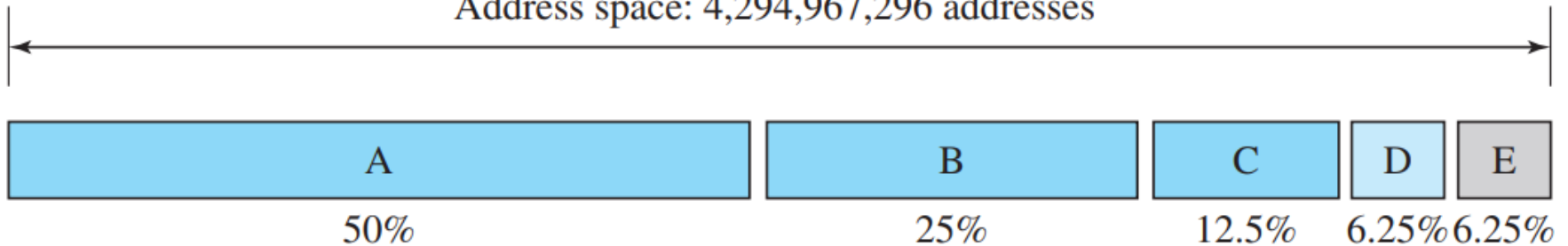


Note

**In classful addressing, the address space is divided into five classes:
A, B, C, D, and E.**

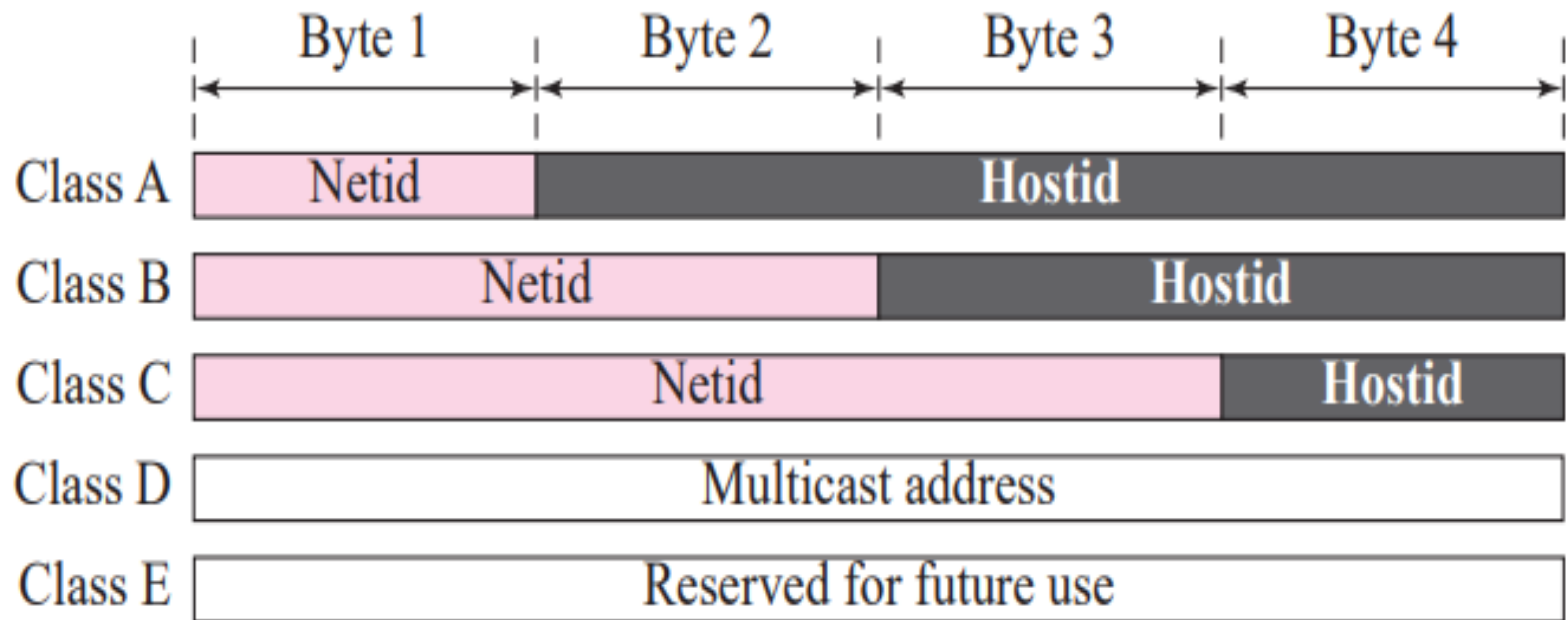
Occupation of address space in classful addressing

Address space: 4,294,967,296 addresses



Class A	0 Prefix	Suffix
Class B	10 Prefix	Suffix
Class C	110 Prefix	Suffix
Class D	1110	Multicast addresses
Class E	1111	Reserved for future use

Class	Prefixes	First byte
A	$n = 8$ bits	0 to 127
B	$n = 16$ bits	128 to 191
C	$n = 24$ bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255





Example 19.4

Find the class of each address.

- a.* 00000001 00001011 00001011 11101111
- b.* 11000001 10000011 00011011 11111111
- c.* 14.23.120.8
- d.* 252.5.15.111

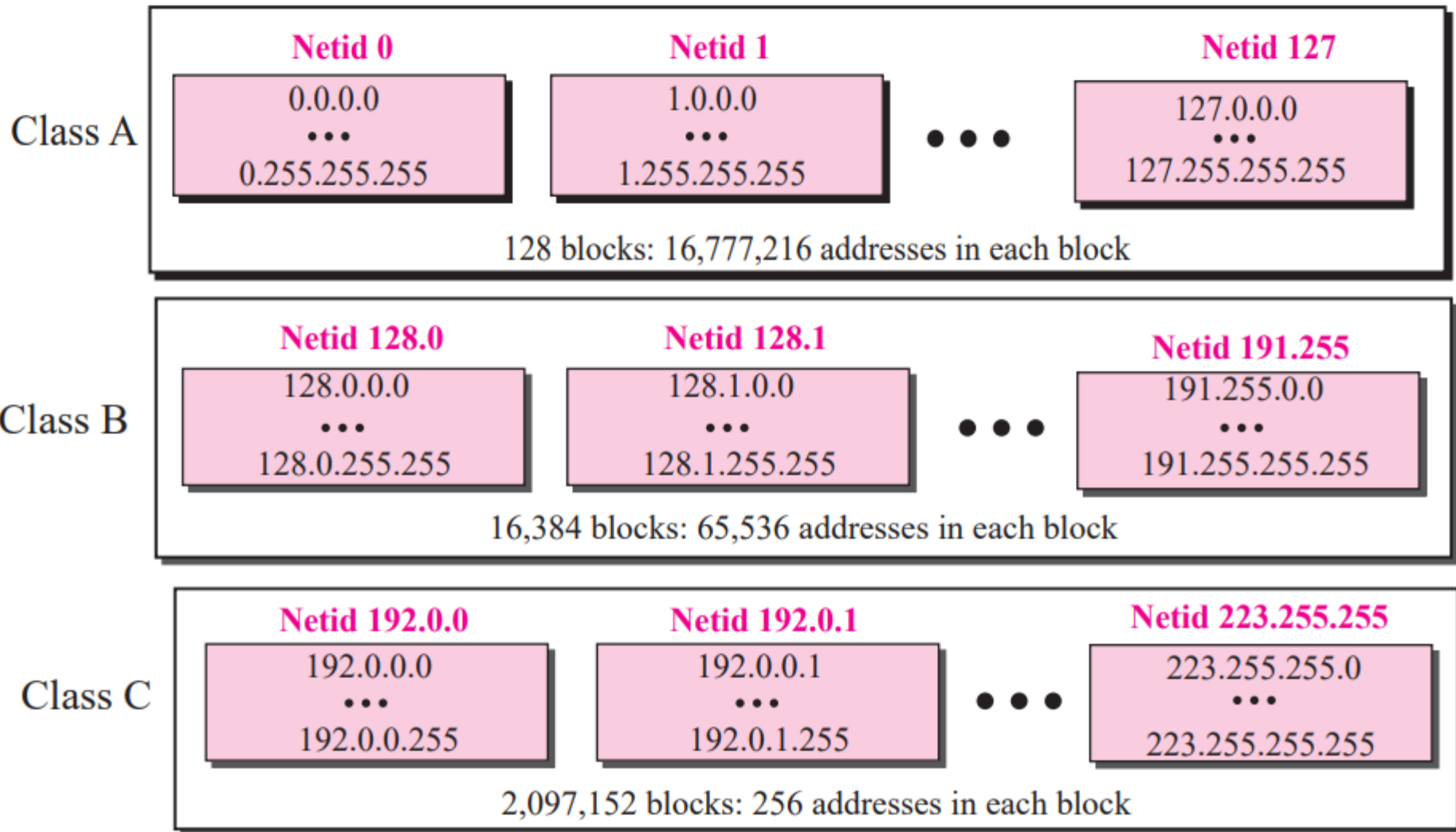
Solution

- a.* The first bit is 0. This is a class A address.
- b.* The first 2 bits are 1; the third bit is 0. This is a class C address.
- c.* The first byte is 14; the class is A.
- d.* The first byte is 252; the class is E.

Table 19.1 *Number of blocks and block size in classful IPv4 addressing*

<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved

Classes and Blocks



Class D

224.0.0.0 ... 239.255.255.255

One block: 268,435,456 addresses

Class E

240.0.0.0 ... 255.255.255.255

One block: 268,435,456 addresses

Note

In classful addressing, a large part of the available addresses were wasted.

Table 19.2 *Default masks for classful addressing*

<i>Class</i>	<i>Binary</i>	<i>Dotted-Decimal</i>	<i>CIDR</i>
A	11111111 00000000 00000000 00000000	255 .0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255 .0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255 .0	/24



Note

In IPv4 addressing, a block of addresses can be defined as

x.y.z.t /n

in which x.y.z.t defines one of the addresses and the */n* defines the mask.



Note

**The number of addresses in the block
can be found by using the formula
 2^{32-n} .**

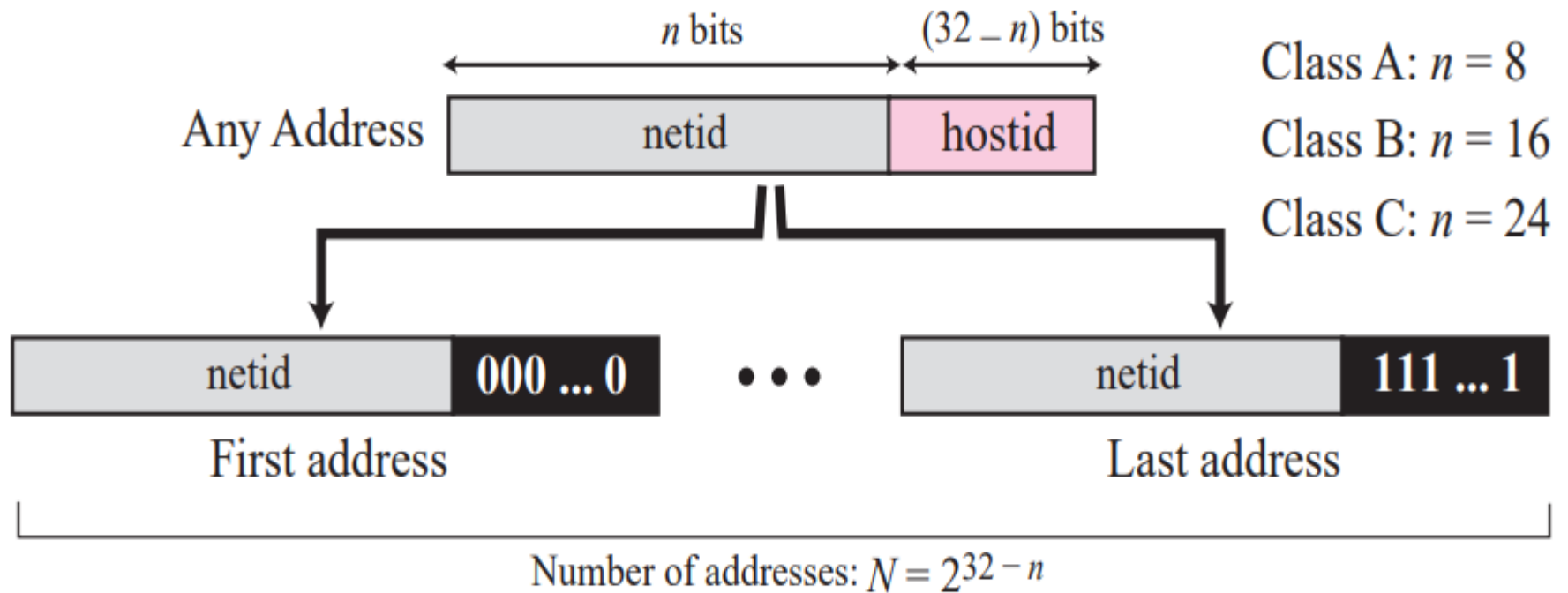
Note

The first address in a block is normally not assigned to any device; it is used as the network address that represents the organization to the rest of the world.

Default mask

A network mask or a default mask in classful addressing is a 32-bit number with n leftmost bits all set to 1s and $(32 - n)$ rightmost bits all set to 0s.

Information extraction in classful addressing



Example : An address in a block is given as **73.22.17.25**. Find the number of addresses in the block, the first address, and the last address.

1.The number of addresses in this block is

$$N = 2^{32-n} = 2^{32-8} = 2^{24} = 16,777,216.$$

2.To find the first address, we keep the leftmost 8 bits and set the rightmost 24 bits all to 0s.

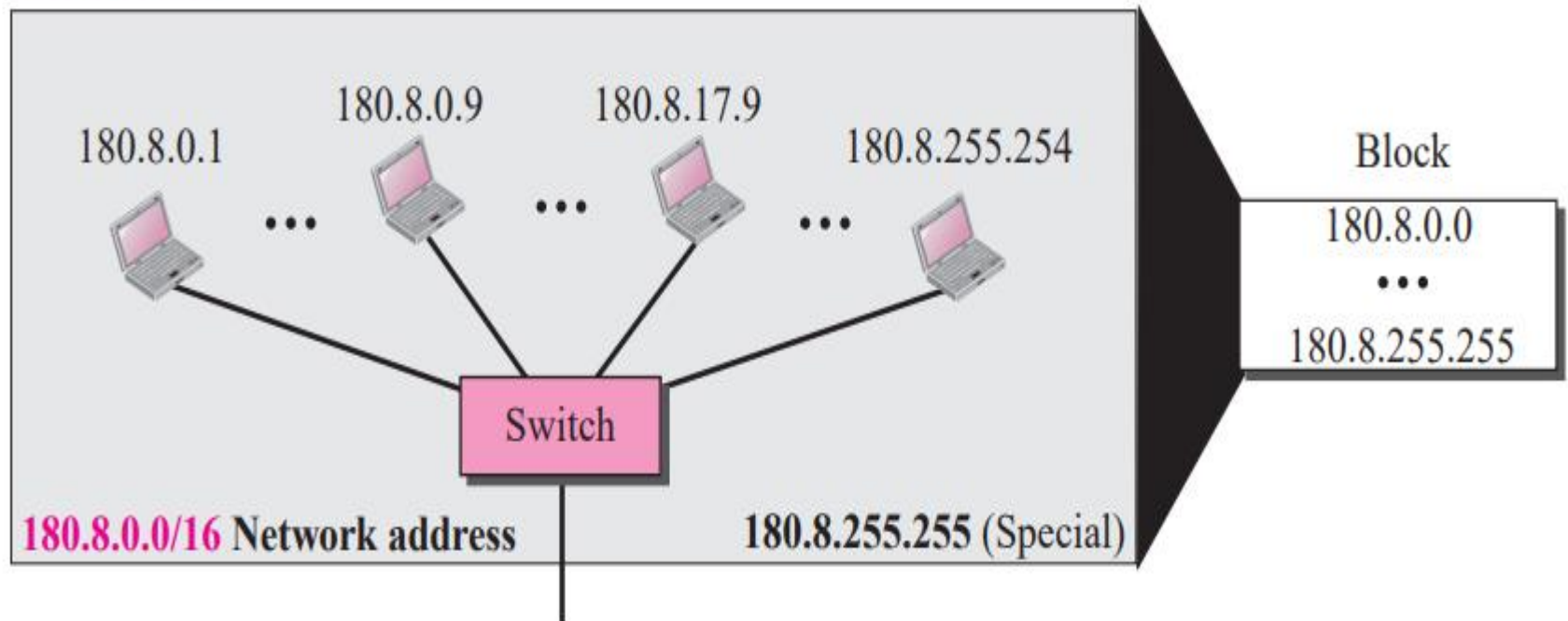
The first address is **73.0.0.0/8** in which 8 is the value of n.

3.To find the **last address**, we keep the leftmost 8 bits and set the rightmost 24 bits all to 1s. The last address is **73.255.255.255**.

Example 2: An address in a block is given as 180.8.17.9. Find the number of addresses in the block, the first address, and the last address.

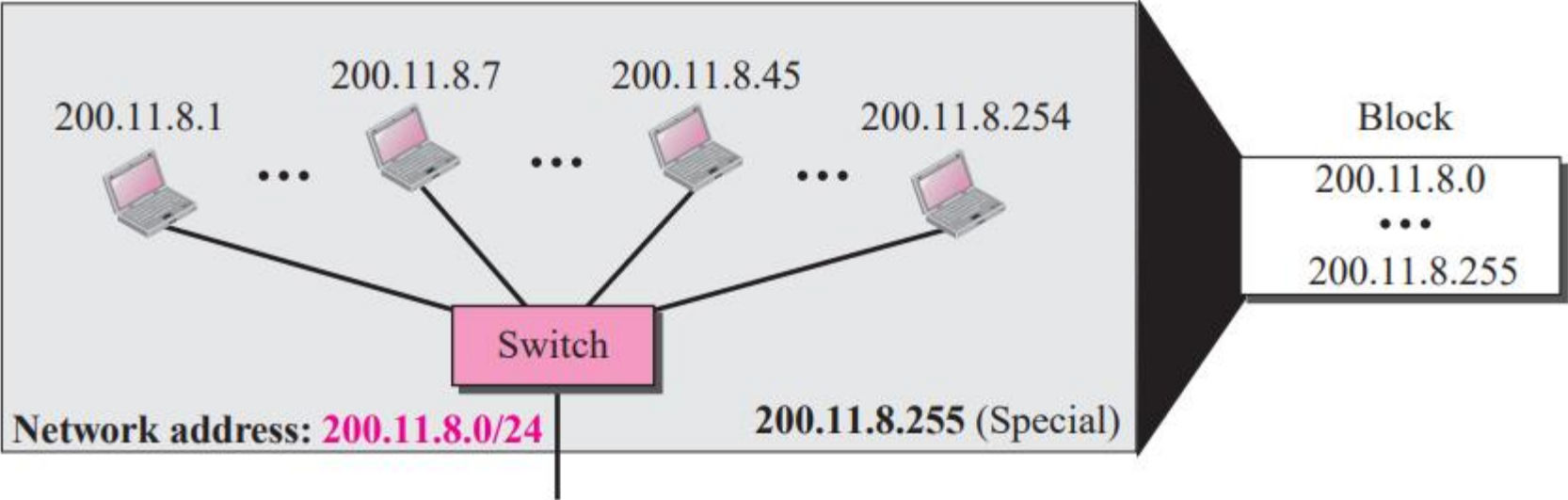
Example 3: An address in a block is given as 200.11.8.45. Find the number of addresses in the block, the first address, and the last address.

Netid 180.8: common in all addresses



The number of addresses in this block is $N = 2^{32-n} = 2^{16} = 65,536$.

Netid 200.11.8: common in all addresses



Scenario:

You are a network administrator for a company that has recently expanded its operations. You need to allocate IP addresses to different departments within the company. The departments and their requirements are as follows:

1. Head office: Given the address 50.17.0.0, allocate IP address for 270 hosts
2. Regional office: Given the address 172.16.10.0, allocate IP address for 100 hosts
3. Branch office: Given the address 192.168.23.0, allocate IP address to 40 hosts
4. Marketing team: Responsible for broadcasting message to all the employees in the company.

Identify the block, mask and range of addresses allocated to each departments



Classful addressing, which is almost obsolete, is replaced with classless addressing.

Classless addressing

- To overcome address depletion and give more organizations access to the internet, classless addressing was designed.
- There are no classes, but the addresses are still granted in blocks.
- The size of the block (the number of addresses) varies based on the nature and size of the entity.
 - Household: 2 addresses
 - Large organization: thousands of addresses.
 - ISP: thousands or hundreds of thousands based on the number of customers it may serve.



In IPv4 addressing, a block of addresses can be defined as

x.y.z.t /n

in which x.y.z.t defines one of the addresses
and the */n* defines the mask.

Note

The first address in the block can be found by setting the rightmost $32 - n$ bits to 0s.



Example 19.6

A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. What is the first address in the block?

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32–28 rightmost bits to 0, we get

11001101 00010000 00100101 00100000

or

205.16.37.32.

This is actually the block shown in Figure 19.3.

Note

The last address in the block can be found by setting the rightmost $32 - n$ bits to 1s.



Example 19.7

Find the last address for the block in Example 19.6.

Solution

The binary representation of the given address is

11001101 00010000 00100101 00100111

If we set 32 – 28 rightmost bits to 1, we get

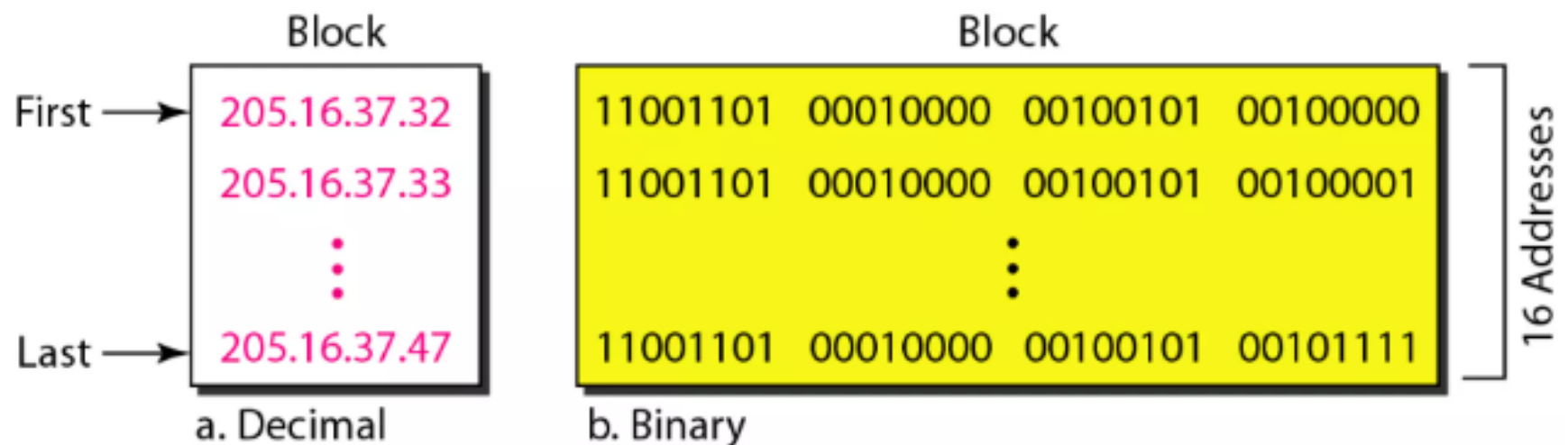
11001101 00010000 00100101 00101111

or

205.16.37.47

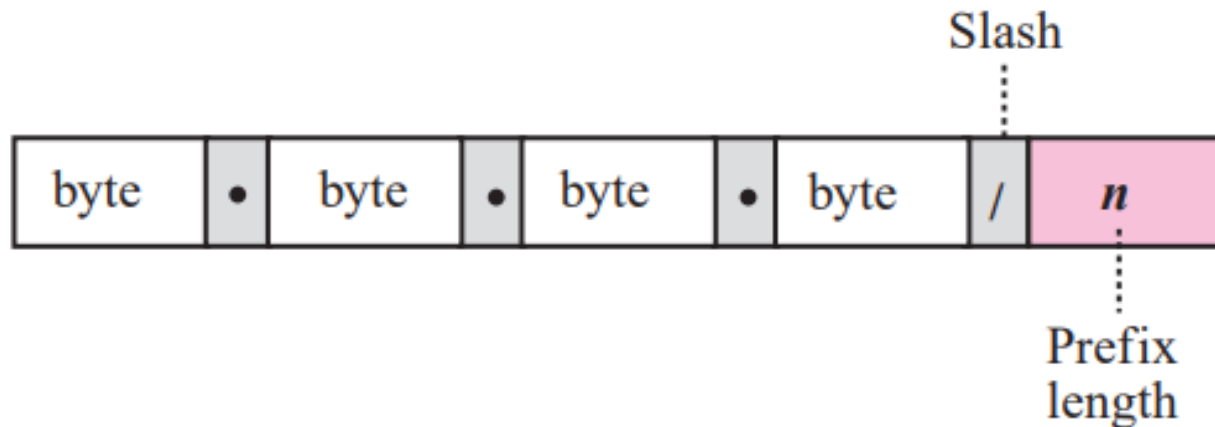
This is actually the block shown in Figure 19.3.

Figure 19.3 *A block of 16 addresses granted to a small organization*



Slash notation

- The slash notation is formally referred to as classless interdomain routing or CIDR notation.



One of the addresses in a block is 167.199.170.82/27. Find the number of addresses in the network, the first address, and the last address.

The number of addresses in the network is
 $= 2^{32-n} = 2^{32-27} = 2^5 = 32$.

We use the AND operation to find the first address (network address). The first address is 167.199.170.64/27.

Address in binary:	10100111 11000111 10101010 01010010
Network mask:	11111111 11111111 11111111 11100000
First address:	10100111 11000111 10101010 01000000

To find the last address, we first find the complement of the network mask and then OR it with the given address: The last address is 167.199.170.95/27.

Address in binary:	10100111	11000111	10101010	01010010
Complement of network mask:	00000000	00000000	00000000	00011111
Last address:	10100111	11000111	10101010	01011111

An organization is granted the block 130.34.12.64/26. The organization needs four subnetworks, each with an equal number of hosts. Design the subnetworks and find the information about each network.

The number of addresses for the whole network can be found as $N = 2^{32-26} = 64$. Using the process described in the previous section, the first address in the network is 130.34.12.64/26 and the last address is 130.34.12.127/26. We now design the subnetworks:

1. We grant 16 addresses for each subnetwork to meet the first requirement (64/16 is a power of 2).
2. The **subnetwork** mask for each subnetwork is:

$$n_1 = n_2 = n_3 = n_4 = n + \log_2 (N/N_i) = 26 + \log_2 4 = 28$$

3. We grant 16 addresses to each subnet starting from the first available address. Figure 5.30 shows the subblock for each subnet. Note that the starting address in each subnetwork is divisible by the number of addresses in that subnetwork.

An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

Solution

There are $2^{32-24} = 256$ addresses in this block. The first address is 14.24.74.0/24; the last address is 14.24.74.255/24. To satisfy the third requirement, we assign addresses to subblocks, starting with the largest and ending with the smallest one.

- a. The number of addresses in the largest subblock, which requires 120 addresses, is not a power of 2. We allocate 128 addresses. The subnet mask for this subnet can be found as $n_1 = 32 - \log_2 128 = 25$. The first address in this block is 14.24.74.0/25; the last address is 14.24.74.127/25.
- b. The number of addresses in the second largest subblock, which requires 60 addresses, is not a power of 2 either. We allocate 64 addresses. The subnet mask for this subnet can be found as $n_2 = 32 - \log_2 64 = 26$. The first address in this block is 14.24.74.128/26; the last address is 14.24.74.191/26.
- c. The number of addresses in the smallest subblock, which requires 10 addresses, is not a power of 2 either. We allocate 16 addresses. The subnet mask for this subnet can be found as $n_3 = 32 - \log_2 16 = 28$. The first address in this block is 14.24.74.192/28; the last address is 14.24.74.207/28.

Figure 20.16 *Format of an IPv6 datagram*

